
PRODUCT REQUIREMENTS DOCUMENT

AI Medication Safety & Visit Prep Platform

Photo-to-Interaction-Check & Longitudinal Health Timeline

Version 1.0 | July 2026 | Project Build Series

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1. Executive Summary

Most people managing an ongoing health condition — especially the elderly — don't see one doctor, they see several: a cardiologist, an endocrinologist, a general physician, maybe a specialist. None of these doctors see what the others have prescribed. As a result, patients accumulate medicines from multiple independent sources with nobody checking whether any of them are dangerous in combination. Layered on top of this, every visit exists in isolation — lab reports and prescriptions stay scattered across time, so a patient walks into each new appointment with no memory of what changed since the last one.

This project is an AI-powered medication safety and visit-preparation platform. A user photographs each new prescription as it arrives. A vision-LLM agent extracts the medicine name, dosage, and frequency, asks a clarifying question when the handwriting or brand name is ambiguous, resolves the Indian brand name to its underlying generic drug, and the system checks the new medicine against everything already on the patient's running list using a versioned drug-interaction knowledge base — surfacing any dangerous combination in plain language, in seconds, to both the patient and a linked family caregiver.

In its second phase, the same running record is extended to lab reports and test results: the system tracks values over time, flags what has genuinely changed, and generates a one-page brief with questions to bring to the next doctor visit — always framed as "discuss this with your doctor," never a diagnosis.

As a student project, this is a production-grade build on the mandated stack: two Dockerised microservices (Express.js + FastAPI with LangGraph), a React frontend, PostgreSQL, custom JWT auth, AWS SES, EC2 deployment with NGINX and HTTPS, CI/CD via GitHub Actions, and OpenTelemetry. Section 9 maps every requirement to a concrete use case.

2. Problem Statement

2.1 Target Users

Patients managing chronic conditions (diabetes, hypertension, thyroid, heart disease) who see 2 or more independent doctors and take 4+ medicines regularly — a population that is large and growing, with India's elderly population alone projected to reach over 220 million by 2036. Their family caregivers, often living in a different city, who are responsible for the patient's wellbeing but have no visibility into day-to-day medicine safety. The platform operator (and, later, a clinical content reviewer): someone who keeps the interaction knowledge base accurate and reviews low-confidence flags before they reach a user.

2.2 Pain Points

- **No one is checking the combination:** Each doctor prescribes in isolation, confident in their own medicine, with no visibility into what the others have already prescribed. Genuinely dangerous overlaps go unnoticed until something goes wrong.
- **Memory resets every visit:** Lab values, prescriptions, and test results live on paper or across disconnected apps. A patient can't easily tell if a number has been quietly trending in the wrong direction over months.
- **Family caregivers are blind:** An adult child living elsewhere has no reliable way to know whether a parent is taking the right medicines correctly, or whether a new prescription conflicts with an existing one — short of constant phone calls.
- **Existing tools solve half the problem:** Reminder apps tell you to take a pill on time but never check if it's safe alongside everything else. Interaction checkers exist but are built for doctors, require manually typing in every drug name, and were never designed to sit quietly in the background of an ordinary person's life.
- **Brand names break the whole chain:** A real Indian prescription says "Dolo 650" or "Glycomet," never "Paracetamol" or "Metformin." Any interaction checker that only understands generic molecule names will silently fail on a real photo, even with perfect OCR, unless something sits in front of it to resolve the brand name first.
- **Appointments are unprepared:** Without a running timeline, patients forget what's changed, forget what to ask, and doctors have to reconstruct history from memory or scattered paperwork each time.

3. Product Vision

"Photograph every prescription and every report — get warned before a dangerous combination reaches you and walk into every appointment already knowing what's changed."

The product combines two connected capabilities: medication safety (photo → structured medicine entry → brand-to-generic resolution → interaction check against a versioned clinical knowledge base → plain-language alert, shared live with a caregiver) and longitudinal visit preparation (report upload → trend detection over time → a one-page, non-diagnostic brief with questions for the next visit). Together they turn a set of disconnected, one-off medical encounters into a single, continuously updating health record.

4. Suggested Product Names

Name	Rationale
MedGuard	Direct, safety-first, easy to say in any language
DawaiDost	"Dawai" (medicine) + "Dost" (friend) — warm, approachable, Hindi-English blend
RxSafe	Short, clinical-sounding, easy as a domain
PillPilot	"Someone's flying the medicine list for you"
HealthLoop	The "one continuous record" framing

5. User Personas

5.1 Persona A — "The Multi-Doctor Patient"

Ramesh, 64, retired, lives in Pune. Manages diabetes and hypertension, sees an endocrinologist and a cardiologist independently, currently on 6 different medicines. His goal: know that a new prescription won't clash with what he's already taking, without having to understand medical jargon himself. His frustration: neither doctor asks what the other has prescribed; he's not confident he'd notice if something felt "off."

5.2 Persona B — "The Long-Distance Caregiver"

Priya, 34, Ramesh's daughter, lives in Bengaluru. Manages her father's health remotely, calls weekly, worries about things she can't see. Her goal: a live view of her father's medicines and any flagged risks, without depending on him to remember and report everything correctly. Her frustration: currently has zero visibility until something goes wrong.

5.3 Persona C — "The Clinical Content Reviewer" (Admin)

Dr. Sana, part-time clinical reviewer for the platform. Reviews prescription extractions the system flagged as low-confidence, checks interaction alerts the system wasn't fully sure about, and maintains the versioned drug-interaction knowledge base and the brand-to-generic mapping table as new data becomes available. Her goal: keep false alarms low enough that users trust the alerts and keep the knowledge base and brand mappings current. Her frustration: needs the system to show its extraction and reasoning clearly, not just a bare verdict, so she can trust or correct it quickly.

6. Core Features

Phase 1 — Medication Safety Guardian

6.1 Prescription Upload & AI Parsing

- Snap-and-upload: user photographs a new prescription (multiple angles encouraged for unclear handwriting) from any phone browser
- Vision extraction: the agent identifies medicine name, dosage, frequency, and prescribing doctor
- Follow-up questions: when handwriting or a brand name is ambiguous, the agent asks one simple clarifying question ("Is this Dolo 650 or Crocin 650?") instead of guessing
- Every extracted medicine is added to one running, always-current list for that patient

6.2 Brand-to-Generic Mapping Layer

- Sits between extraction and the interaction check: every extracted medicine name (a brand name, in almost all real cases) is resolved to its generic/molecule name before any interaction lookup runs
- Backed by a versioned brand→generic table, seeded from an open Indian medicine dataset (brand name, generic name, composition)
- If a brand name can't be confidently resolved, the medicine is added to the list with a "generic unresolved" flag and routed to the admin review queue rather than silently skipping the interaction check
- This layer is treated as at least as important as the interaction database itself — it's the step most likely to silently break on a real prescription photo

6.3 Interaction Safety Engine

- A versioned drug-interaction knowledge base (curated subset built from an open, freely licensed interaction dataset) is checked deterministically every time a new (resolved-to-generic) medicine is added, against everything already on the list
- Severity levels (e.g. avoid combination, monitor closely, minor/no action) shown with a plain-language explanation of why — never just a red flag with no reasoning
- Quotes its own confidence: high-confidence flags go straight to the user; low-confidence or borderline cases are queued for the clinical reviewer before being shown

6.4 Caregiver Linking & Shared View

- Patient can invite a family member by email; the caregiver gets a linked, permissioned view of the same medicine list and any active alerts

- Caregiver role is read-plus-acknowledge (can see and confirm they've seen an alert, cannot edit the medical record directly)

6.5 Notifications

- SES-powered alerts the moment a new interaction is flagged
- Confirmation email whenever a new medicine is successfully added
- A simple weekly summary of the current medicine list, sent to both patient and caregiver

Phase 2 — Longitudinal Visit Prep (Extension)

6.6 Lab Report Timeline

- Upload lab reports and test results over time (same photo-upload pattern as prescriptions)
- Structured extraction of test values, matched against known test types
- Deterministic trend calculation against the patient's own history — flags a value only when it has moved meaningfully, not on every small fluctuation

6.7 Visit Prep Brief Generator

- Ahead of a logged upcoming appointment (or on demand), generates a one-page brief: what's changed since the last visit, the current medicine list, and 3–4 specific questions to ask
- Every brief end with the same explicit framing: "Discuss this with your doctor — this is not a diagnosis."

Shared / Platform Features

6.8 Admin & Clinical Review Dashboard

- Review queue for low-confidence prescription extractions, unresolved brand names, and borderline interaction flags, with the AI's extracted data and reasoning shown side by side
- Interaction knowledge base and brand-mapping table management, both versioned with effective dates so every historical alert stays explainable
- Basic usage and alert-volume stats

6.9 Awareness Layer

- Every interaction alert explains, in plain language, why the combination is risky (e.g. "both of these affects how your blood clots — ask your doctor if you still need both") — building health literacy over time rather than just issuing warnings

6.10 Consent & Data Privacy (DPDP-Aligned)

- Affirmative, explicit opt-in consent at signup for the collection and processing of health data — no pre-ticked boxes, no bundled/blanket consent
- A short, plain-language notice (not a legal wall of text) explaining what's collected, why, and who can see it (including what a linked caregiver can and can't see)
- A working "delete my record" path that a patient can trigger themselves, with a clear explanation of what gets deleted vs. retained (e.g. anonymized audit logs)
- Consent and deletion actions are themselves logged (who consented to what, when) so the platform can demonstrate compliance, not just claim it

7. User Journey

Phase	User Action	System Response
1	Ramesh photographs a new prescription from his cardiologist	Vision agent extracts the brand name and dosage, resolves it to its generic name via the mapping layer
2	Answers one follow-up: "Is the dosage clearly 5mg?" — Yes	Medicine added to his running list, generic name confirmed
3	System checks the new medicine against his existing 5 medicines	Flags a moderate interaction with a medicine his endocrinologist prescribed, with a plain-language explanation
4	Priya (linked caregiver) receives the same alert by email	She calls her father and both bring it up at the next appointment
5	A few weeks later, Ramesh uploads a new blood sugar report	System extracts the value and compares it to his last two reports — flags a rising trend
6	Ramesh has a cardiology appointment logged for next week	System generates a one-page brief: the flagged interaction, the rising sugar trend, and 4 questions to ask
7	Ramesh brings the brief to his appointment	Doctor confirms the interaction is worth adjusting; Ramesh updates his list afterward
8	Dr. Sana checks the admin dashboard	1 low-confidence extraction reviewed and corrected; a new brand name added to the mapping table

8. Competitive Landscape

Capability	This Product	Generic Reminder Apps	1mg / PharmEasy	Clinical Interaction Checkers
Photo-based prescription capture	Yes	No (manual entry)	No	No
Understands Indian brand names, not just generics	Yes	N/A	Yes (catalogue)	No (generic name entry only)
Checks your actual running list for interactions	Yes	No	Per-drug lookup only, not your list	Yes, but manual entry of every pair
Built for patients & families, not just clinicians	Yes	Yes	Yes	No
Caregiver-shared live view	Yes	Rarely	No	No
Tracks lab trends over time and preps visit questions	Yes	No	No	No
Plain-language "why," not just a flag	Yes	N/A	Sometimes	No

Interaction checkers have existed for years — but they're built for clinicians typing in known drug pairs, not for a family photographing prescription as they arrive and getting warned automatically, in a language the prescription was actually written in (a brand name). That gap — patient-facing, photo-driven, brand-name-aware, always-running-in-the-background safety — is the hole this product fills.

9. Technical Requirements & Use Cases

The following requirements are mandatory (except the two marked optional). The right-hand column shows where each fit in this product.

#	Requirement	Suggested Use Case in This Project
1	GitHub Actions for CI/CD	On push to main: lint + tests for ms1, ms2, frontend; build and push Docker images; SSH-deploy to EC2 and restart containers. PR checks block merge; LLM/AWS keys stored in GitHub Secrets.
2	Single GitHub monorepo (ms1, ms2, frontend)	/ms1-core-api (Express), /ms2-agent-service (FastAPI + LangGraph), /frontend (React), /infra (NGINX config, docker-compose, deploy scripts). Path-filtered workflows rebuild only what changed.
3	AWS SES for email	Interaction-alert emails, new-medicine confirmations, weekly medicine summaries, caregiver invite emails, visit-prep brief delivery, admin alerts for low-confidence flags.
4	React for frontend	Patient/caregiver app: prescription/report upload (camera or file), follow-up Q&A, medicine list view, alert timeline, visit-brief screen. Separate admin dashboard views for the clinical reviewer.

#	Requirement	Suggested Use Case in This Project
5	At least one PostgreSQL database	Stores users, caregiver links (with permission level), medicines (current + historical), brand→generic mappings, interaction flags, lab reports, lab values, visits, generated briefs, and a versioned interaction knowledge base table.
6	Docker for both microservices (+ router)	Dockerfile per service; ms2's image includes image-preprocessing dependencies. NGINX acts as the router: public traffic hits ms1 and the frontend; ms2 is reachable only internally over the Docker network, never exposed publicly.
7	Custom JWT auth (Google OAuth a plus)	Email/password with bcrypt and refresh tokens; role claims: patient / caregiver / admin. Patients see only their own record; caregivers see only records they're linked to; admins see the review queue and knowledge base. Google OAuth as a bonus login option.
8	Domain integration everywhere	medguard.in for the app, api.medguard.in for ms1; ms2 stays on an internal route, not a public subdomain. SES sends from care@medguard.in with SPF/DKIM verified so alert emails don't land in spam.
9	LangGraph / LangChain in at least one microservice	ms2 is the dedicated agent service, with three LangGraph graphs: (a) Prescription assessment graph, (b) Lab trend graph, (c) Visit-brief writer graph.
10	At least two microservices — Express.js + FastAPI	ms1 (Express.js): auth/roles, patient and caregiver records, medicine list, alerts, visits, SES dispatch, admin dashboard data. ms2 (FastAPI): image handling and all three LangGraph graphs.
11	AWS EC2 deployment with HTTPS via certbot	All containers on one EC2 instance via docker-compose; certbot with auto-renew. Free-tier note: a t2.micro/t3.micro instance on the AWS Free Tier (750 hrs/month for 12 months) is enough for an MVP demo.
12	NGINX as reverse proxy	TLS termination and routing: / → React build, /api → ms1; ms2 has no public route at all. Raise client_max_body_size for phone photo uploads (3–8 MB each).
13	Observability with OpenTelemetry	Trace the medicine-add pipeline: upload → vision call → follow-up loop → brand resolution → interaction lookup → alert dispatch, propagated across ms1 → ms2. Free option: self-host Jaeger via docker-compose on the same EC2 box.
14	Modular, clean, readable code	Layered structure per service; the interaction knowledge base, brand-mapping table, and severity rules live as versioned data modules, never hard-coded into prompts; the deterministic interaction-lookup logic lives in one pure, tested module.
15	(Optional) API documentation	FastAPI's built-in OpenAPI docs for ms2; a Postman collection or swagger-jsdoc for ms1, including the medicine-add and alert lifecycle and the ms1 ↔ ms2 contract.
16	(Optional) Rate limiting	Caps on assessment requests per user per hour (controls LLM cost on a public-facing upload endpoint), login attempt limits, and upload size limits, on top of a global NGINX rate limit.

10. System Architecture

10.1 Components

- Frontend (React): upload flow (prescription/report photo → follow-up Q&A → medicine list / alert timeline / visit-brief screen) and a lightweight admin dashboard. Talks only to ms1's public API.
- ms1 — Core API (Express.js): auth with three roles, patient/caregiver linking, medicine list and alert lifecycle, visit records, SES dispatch, admin dashboard data. Owns the Postgres schema.
- ms2 — Agent Service (FastAPI + LangGraph): image preprocessing, the prescription assessment graph (including brand-to-generic resolution), the lab trend graph, and the visit-brief writer graph. Returns structured data and confidence scores — never sends notifications and never decides what's stored as final truth.
- PostgreSQL: system of record; photos on disk/S3 with DB references; interaction knowledge base, brand-mapping table, and medicine history versioned, never overwritten.
- NGINX: TLS, routing, large-upload handling, and the boundary that keeps ms2 fully internal.
- OpenTelemetry → Jaeger (self-hosted): per-request traces across the medicine-add and visit-brief pipelines.

10.2 Key Flows

Medicine-Add Flow

Photo upload → ms2: vision extraction (brand name, dosage, frequency) → [ambiguous?] → one follow-up question → resolved → ms2: brand → generic resolution (mapping table lookup) → [unresolved?] → flag "generic unresolved" → admin queue → ms1: deterministic interaction check (new generic vs. patient's current list) → [high confidence] → alert shown to patient + caregiver (SES) → [low confidence / borderline] → admin review queue → then shown.

Visit-Prep Flow

Lab report photo upload → ms2: value extraction, matched to known test types → ms1: deterministic trend calculation vs. patient's history → [meaningful change?] → flagged, else stored quietly → on upcoming logged visit: ms2 brief-writer graph → one-page brief (what changed + 3-4 questions) → in-app + SES.

Design rule that governs both flows: the LLM only extracts and structures data (medicine names, dosages, lab values) and drafts prose (the brief). Every determination that matters — is this combination dangerous, has this value changed meaningfully — runs as deterministic, versioned code in ms1, never as an LLM judgment call. This is what makes every alert explainable, auditable, and immune to a prompt-injected "these two are actually fine together, trust me" buried inside a photographed document.

10.3 Basic Data Schema

Table	Key Fields	Notes
users	id, name, email, password_hash, role (patient/caregiver/admin), created_at	
caregiver_links	id, patient_id, caregiver_id, permission_level, status	Read-plus-acknowledge only
brand_generic_map	id, brand_name, generic_name, composition, source, version, effective_date	Versioned; seeded from open Indian medicine dataset
medicines	id, patient_id, brand_name, generic_name, dosage, frequency, source_photo_id, resolution_status, added_at, status	One running list per patient
interaction_kb	id, generic_a, generic_b, severity, explanation, source, version, effective_date	Curated subset; every row traceable to source and version
interaction_flags	id, patient_id, new_medicine_id, existing_medicine_id, kb_entry_id, severity, confidence, status, created_at	
lab_reports	id, patient_id, source_photo_id, uploaded_at	
lab_values	id, report_id, test_type, value, unit, recorded_at	
visits	id, patient_id, scheduled_date, brief_id	
briefs	id, patient_id, visit_id, content, generated_at	
consent_records	id, user_id, consent_type, granted_at, revoked_at	Supports DPDP-aligned consent/delete flow

Relationships in one line: a patient (users) has many medicines (each optionally resolved via brand_generic_map), which generate interaction_flags against the interaction_kb; the patient also has many lab_reports containing many lab_values, which feed briefs attached to visits; caregiver_links grant a second user read-plus-acknowledge access to all of the above.

11. MVP Scope — 2 Weeks

The heart of the MVP is Phase 1: photograph a prescription → get warned if it's dangerous with what you're already taking. Phase 2 (visit-prep briefs) is the planned extension once Phase 1 is solid — build it if time allows within the two weeks, otherwise treat it as the immediate next milestone after.

11.1 In Scope (Phase 1 — MVP)

- Photo upload (1–3 photos) → prescription assessment graph → structured medicine entry with confidence (one follow-up question maximum at MVP)
- A hand-built brand-to-generic mapping subset (roughly 50–150 common brand names) for the MVP medicine set, sourced from an open Indian medicine dataset
- Versioned interaction knowledge base for a limited, well-covered set of common chronic-condition medicines (diabetes, hypertension, cholesterol), hand-curated as a subset of a larger open dataset (roughly 50–150 pairs)
- Deterministic interaction check on every new medicine against the patient's current list, with plain-language severity and reasoning
- Caregiver invite + linked, permissioned shared view
- SES alerts: new medicine confirmation, interaction flag, weekly summary
- Admin review queue for low-confidence extractions, unresolved brand names, and borderline flags
- A basic consent screen at signup (plain-language notice + affirmative opt-in) and a working delete-my-record action
- Three-role JWT auth (patient / caregiver / admin), full mandatory infra: monorepo, Docker, CI/CD, EC2 + NGINX + certbot + domain, OTel traces on the medicine-add pipeline

11.2 Stretch Within the Two Weeks / Immediate Next Milestone

Lab report upload and trend detection (6.6), visit-brief generation (6.7), Google OAuth, rate limiting, and API docs.

11.3 Two-Week Plan

Days	Milestone	Proof It Works
1–2	Monorepo scaffold, Docker + compose (ms1, ms2, Postgres, NGINX), schema v1, CI pipeline; vision-LLM spike on 10 real prescription photos	Stack boots with one command; the model extracts medicine names reliably
3–4	JWT auth (patient/caregiver/admin); deploy skeleton to EC2 with HTTPS + domain; SES verification started; interaction KB and brand-mapping subset drafted; consent screen built	Login and photo upload work from a phone on the live domain
5–7	Assessment graph in ms2: vision → follow-up (≤ 1) → brand resolution →	An uploaded prescription photo returns a structured entry and, if relevant, a flagged interaction

Days	Milestone	Proof It Works
	structured entry + confidence; interaction-lookup module in ms1 (pure, tested); alert screen with plain- language reasoning	
8–9	Caregiver invite + linked view; SES alert and confirmation emails	Inviting a caregiver produces a working linked account that sees the same alerts
10–11	Admin dashboard: review queue with AI reasoning shown, knowledge base and brand-mapping editor with versioning	Admin corrects a low-confidence extraction and it's reflected system-wide
12	OTel traces across upload → vision → brand resolution → interaction-check → alert; cost-per-check visible	One trace spans the full medicine-add pipeline with cost attributes
13–14	Hardening: irrelevant/bad photo rejection, EXIF stripping, upload limits, empty states; final deploy, README + demo video; start on Phase 2 if time remains	End-to-end demo: photo → structured medicine → interaction check → alert → caregiver notified

12. Learning Outcomes

- Building a vision-LLM product on sensitive personal data: image preprocessing, multimodal prompting, follow-up loops for ambiguity, confidence-aware outputs
- The LLM-proposes / code-decides pattern applied to a safety-critical domain: separating probabilistic extraction from deterministic, auditable interaction checks — the most important lesson in this project
- Designing versioned reference data (an interaction knowledge base and a brand-to-generic mapping table) so every historical alert stays explainable
- Handling real-world naming ambiguity: normalizing brand names to generics before any safety-critical logic runs, and deciding what to do when that normalization fails
- Modelling a multi-role system (patient / caregiver / admin) with permissioned, shared access to the same underlying record
- Handling user-generated medical images safely: size limits, EXIF/location stripping, irrelevant-content rejection on a public endpoint
- Designing a real (not decorative) consent and deletion flow for health data

- Shipping production infrastructure on a free-tier budget: Docker, NGINX, HTTPS, domains, CI/CD, SES, and OpenTelemetry with per-check cost telemetry

13. Evaluation Criteria

Area	Weight	What Evaluators Look For
Core product loop	25%	Photo → extracted medicine → brand resolution → interaction check → plain-language alert → caregiver notified, all live from a phone
Agentic quality (ms2)	20%	LangGraph used meaningfully: extraction → follow-up → brand resolution → structured-entry graph with confidence; sensible behaviour on ambiguous and irrelevant photos; LLM kept out of the final interaction verdict
Architecture & code quality	20%	Clean ms1/ms2 boundaries; interaction-lookup as a pure tested module; versioned knowledge base and brand-mapping table; typed schemas; no comment clutter
Infrastructure & deployment	20%	Live on the domain with HTTPS; mobile camera upload works; Docker, CI/CD, SES flows, NGINX upload config, and OTel traces with cost attributes all demonstrable
Security & correctness	10%	Role separation, EXIF stripping, upload validation, hashed passwords, no secrets in repo, every alert traceable to the knowledge base version that produced it, working consent/delete flow
Extras	5%	API docs, rate limiting, Phase 2 (visit-prep) progress, tests

14. Stretch Goals

- Phase 2 fully built out: lab trend detection + visit-prep brief generator
- WhatsApp entry point: send a prescription photo on WhatsApp → get the check result back
- Generic/cheaper-alternative suggestion when a branded medicine is added (referencing publicly available generic substitute data)
- Multi-language alerts and briefs (Hindi and regional languages)
- Household Health Risk Cascade (Phase 3 direction): extend the shared record across a whole family so a new diagnosis in one member prompts proactive screening reminders for at-risk relatives
- Voice-note input as an alternative to typing for less tech-comfortable users
- Full brand-mapping coverage beyond the MVP subset, moving from a hand-curated table to a fuller catalogue-backed lookup

15. Data Sources & Licensing

Sourced to unblock the two knowledge bases the product depends on (drug interactions, and brand-to-generic mapping) plus benchmarking data for the extraction step.

Dataset	Use	Licensing Note
DDInter	Primary source for the interaction knowledge base (6.3) — curated, 240,000+ drug-drug pairs with severity classifications, freely available	No registration or license negotiation needed
TWOSIDES	Supplementary signal only — statistical drug-pair/adverse-event associations mined from FDA reports	Open, but not clinically curated — treat as a supplement, never a replacement for DDInter
DrugBank	Spot-checking entries by hand only, not for the actual knowledge base	Requires a paid license for most use; free tier only for limited academic cases
Kaggle: mohneesh7/indian-medicine-data	Seeds the brand→generic mapping table (6.2)	Usable as-is for a student build
1mg catalogue scrape (~300K rows)	Optional larger source for brand/generic/composition if the Kaggle set proves too small	Licensed CC BY-NC-SA — fine for a student build, not for commercial use without renegotiating
MIRAGE	Benchmarking OCR/extraction accuracy — built specifically for multimodal recognition of Indian prescriptions	Closest match to the actual use case; use for held-out evaluation
RxHandBD	General handwriting sanity check — 5,578 cropped handwritten prescription words, 80/20 split	Useful baseline, not India-specific
Kaggle illegible-prescription set / IEEE DataPort handwritten-vs-printed set	Stress-testing the "ask a follow-up question instead of guessing" behavior	Use for qualitative review, not primary accuracy metric

Scope decision for the MVP: given diabetes, hypertension, and cholesterol are the target drug classes, both the interaction knowledge base and the brand-mapping table are hand-curated subsets (roughly 50–150 entries each) rather than the full source datasets — this keeps both stores small enough to version, explain, and demo confidently to an evaluator.

16. Open Questions — Researched Answers

Each original open question is restated, followed by what current research and prevailing practice suggest for a two-week student build, plus what to revisit if the project ever moves past a demo.

16.1 Knowledge Base Sourcing

Status: resolved in v1.1, unchanged. DDInter remains the right primary source (open, severity-labeled, no license negotiation), TWOSIDES stays supplementary-only, and DrugBank stays spot-check-only given its paid-license terms. The one durable caveat is the 1mg scrape's CC BY-NC-SA license, which blocks commercial use without a separate agreement — not a blocker for a student demo, but flag it in the README so a future team doesn't accidentally ship it into a commercial fork.

16.2 Brand-Mapping Data Quality

No independent audit of the Kaggle indian-medicine-data set's accuracy exists publicly, so the review process has to be built into the product rather than assumed away. Recommended process:

- Every row imported from the Kaggle set is marked source: kaggle_seed, version: v1, effective_date: <import date> in brand_generic_map — never silently overwritten.
- Any brand name that resolves through the seed table but has never been confirmed by a human is shown to Dr. Sana's review queue the first time it's actually used in a real interaction check that produces a non-trivial (moderate/major) flag.
- When Dr. Sana corrects or confirms a mapping, it's written as a new versioned row (source: reviewer_confirmed, version: v2) rather than an edit-in-place, so any interaction flag generated under the old mapping stays explainable.
- For MVP scope, a lightweight manual spot-checks of all ~50–150 seeded rows against a second source before demo day is enough.

16.3 OCR/Extraction Confidence Threshold

There's no single "correct" number here — production confidence-routing practice converges on a few consistent design principles rather than one threshold:

- Use tiered routing, not a single cutoff: high confidence → shown directly; a middle band → shown to the user but also queued for background review; low confidence → held for review before the user sees it at all.
- Calibrate per field, not per document. Medicine name and dosage (safety-critical) should have a stricter threshold than prescribing-doctor name (cosmetic).

- For this MVP: start conservative (e.g. route anything below ~85% confidence to the admin queue), then watch the review-queue composition during the demo week to tune the threshold.
- Because this is a safety-critical use case, the system should default toward more review rather than less — an unresolved brand name or a low-confidence dosage should never reach the interaction check silently.

16.4 Non-Diagnostic Framing / Disclaimer Language

The product's framing rule — "Discuss this with your doctor — this is not a diagnosis" — should appear consistently rather than being invented per-screen. Concrete guidance:

- State what the system did (extracted, matched, flagged a pattern) rather than what it concluded about the patient's health.
- Keep the disclaimer attached to the specific claim it qualifies, not just buried in a footer or terms-of-service page.
- Avoid words that imply clinical judgment about the individual (e.g. "you should," "your condition requires") — favor "ask your doctor whether," "worth discussing," "flagged for review."
- Make the disclaimer identical, or near identical, wherever it appears so it reads as a stable product policy rather than a one-off legal hedge.

16.5 Data Privacy — DPDP Act Status as of July 2026

The DPDP Rules, 2025 were notified on 13 November 2025, and the Data Protection Board of India is now operational. The Act is formally "in force," but most substantive day-to-day obligations become enforceable on 13 May 2027 (18 months from the Rules' notification), not immediately. 2026 is explicitly the "build year" for compliance readiness, not the enforcement year.

Key points for the project:

- Breach notification: the Rules require notifying the Data Protection Board without delay, with a detailed follow-up report within 72 hours, and notifying affected individuals in clear, accessible language. There is currently no materiality threshold in the statutory text.
- Significant Data Fiduciary (SDF) status: SDF designation is not self-assessed — it's a formal Central Government notification under Section 10. A two-week student build at demo scale would not plausibly be designated.
- Data localization: the Rules give the government power to require that specified categories of personal data never leave India, but the specific list has not yet been notified. The practical mitigation is to keep patient data and backups on India-region infrastructure from day one (AWS ap-south-1).

Net effect on scope: the MVP's plain-language notice, affirmative opt-in, working delete path, and consent/deletion audit log (6.10) remain the right MVP-scope answer. The main addition worth carrying forward: log breach-relevant events (who accessed what, when) with the same rigor as consent events.

16.6 Caregiver Permission Granularity

The PRD's existing "read-plus-acknowledge" role is a reasonable default, but a small number of concrete tiers covers the realistic cases without over-building:

- Full view (default): sees the complete medicine list, all interaction alerts, and lab trends — this is what most long-distance caregivers actually want.
- Alerts-only: sees only active interaction flags and their plain-language explanation, not the full medicine list or lab values — useful for a secondary caregiver.
- The patient, not the platform, chooses the tier per caregiver at invite time and can change it later.

For the two-week MVP, shipping only the Full view tier is a defensible scope cut — document it as a known simplification.

16.7 Cost Control

- Rate-limit at the point of the first LLM call (image upload → vision extraction), not just at the follow-up-question step, since that first call is the expensive one regardless of how the request resolves afterward.
- Track and expose the per-check cost as an OpenTelemetry span attribute — this is what actually lets you set a sane rate limit instead of guessing.
- Reject obviously invalid uploads (wrong file type, no image content, duplicate photo of the same prescription re-submitted) before they reach the vision model at all.
- A simple, generous starting cap (e.g. a low double-digit number of assessment requests per user per hour) is enough for a demo.

MedGuard

AI Medication Safety & Visit Prep Platform

Sarvesh Chandran · Sahil · Nyasha Das

Version 1.0 | July 2026

